

2.4 Integration

Further Pure 2 · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Integration may require substitution, partial fractions, integration by parts or reduction formulae. Check domains for logarithmic antiderivatives; $\int 1/x \, dx = \ln|x| + C$. Definite integrals can represent signed area, so use absolute value or split where actual geometric area is asked.

Worked example: follow the reasoning

$\int x e^x \, dx = x e^x - \int e^x dx = e^x(x-1) + C$ by parts.

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Evaluate $\int$ from 0 to 1 of $x e^x \, dx$.

[3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

Parts gives antiderivative $e^x(x-1)$ [2]; evaluate at 1 minus at 0: $0 - (-1) = 1$ [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Further Mathematics 9231, 2026–2027.

<https://www.cambridgeinternational.org/Images/697357-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.